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Crown bulk for FinFET device

Abstract

A method includes forming an anti-punch-through layer over a first region and a second region of a substrate, forming a semiconductor layer over the anti-punch-through layer, patterning the semiconductor layer and the anti-punch-through layer to form a first plurality of fins over the first region and a second plurality of fins over the second region, and forming a patterned resist layer over the first plurality of fins and the second plurality of fins. The method also includes recessing a portion of the substrate between the first plurality of fins and the second plurality of fins in an etching process through openings of the patterned resist layer.

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Background/Summary

PRIORITY DATA (1) This is a divisional application of U.S. patent application Ser. No. 16/827,315 filed Mar. 23, 2020, which claims the benefit of U.S. Patent Provisional Application No. 62/907,258 filed Sep. 27, 2019, entitled “Crown Bulk in FinFET Structure for Isolation and Latch Up Improvement”, the entire disclosures of which are incorporated herein by reference.

BACKGROUND

(1) The semiconductor integrated circuit (IC) industry has experienced exponential growth. Technological advances in IC materials and design have produced generations of ICs where each generation has smaller and more complex circuits than the previous generation. In the course of IC

evolution, functional density (i.e., the number of interconnected devices per chip area) has generally increased while geometry size (i.e., the smallest component (or line) that can be created using a fabrication process) has decreased. This scaling down process generally provides benefits by increasing production efficiency and lowering associated costs. Such scaling down has also increased the complexity of IC structures (such as three-dimensional transistors) and processing and, for these advancements to be realized, similar developments in IC processing and manufacturing are needed. For example, device performance (such as device performance degradation associated with various defects) and fabrication cost of field-effect transistors become more challenging when device sizes continue to decrease. Although methods for addressing such a challenge have been generally adequate, they have not been entirely satisfactory in all aspects.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) Aspects of the present disclosure are best understood from the following detailed description when read with the accompanying figures. It is noted that, in accordance with the standard practice in the industry, various features are not drawn to scale. In fact, the dimensions of the various features may be arbitrarily increased or reduced for clarity of discussion.
- (2) FIGS. 1A, 1B, 1C, 1D, 1E, and 1F illustrate a process for fabricating finFET structures with a crown bulk, according to one example of principles described herein.
- (3) FIG. 2 is a diagram showing an illustrative image of finFET structures with a crown bulk, according to one example of principles described herein.
- (4) FIG. 3 is a flowchart showing an illustrative method for forming finFET structures with a crown bulk, according to one example of principles described herein.

DETAILED DESCRIPTION

- (5) The following disclosure provides many different embodiments, or examples, for implementing different features of the provided subject matter. Specific examples of components and arrangements are described below to simplify the present disclosure. These are, of course, merely examples and are not intended to be limiting. For example, the formation of a first feature over or on a second feature in the description that follows may include embodiments in which the first and second features are formed in direct contact, and may also include embodiments in which additional features may be formed between the first and second features, such that the first and second features may not be in direct contact. In addition, the present disclosure may repeat reference numerals and/or letters in the various examples. This repetition is for the purpose of simplicity and clarity and does not in itself dictate a relationship between the various embodiments and/or configurations discussed.
- (6) Further, spatially relative terms, such as “beneath,” “below,” “lower,” “above,” “upper” and the like, may be used herein for ease of description to describe one element or feature's relationship to another element(s) or feature(s) as illustrated in the figures. The spatially relative terms are intended to encompass different orientations of the device in use or operation in addition to the orientation depicted in the figures. The apparatus may be otherwise oriented (rotated 90 degrees or at other orientations) and the spatially relative descriptors used herein may likewise be interpreted accordingly.
- (7) The present disclosure is generally related to semiconductor devices and the fabrication thereof, and more particularly to methods of fabricating field-effect transistors (FETs), such as fin-like FETs (FinFETs). A finFET transistor typically includes a fin-like semiconductor structure formed on a substrate. The substrate and the bottom portion of the fin-like structure may be doped to form a doped well. For example, for an NMOS transistor, a p-well may be formed within the lower portion of the fin structure. For a PMOS transistor, an n-well may be formed in the lower portion of

the fin structure.

(8) Transistors are used to form complex logic circuits within integrated circuits. For these circuits, p-type (PMOS) transistors may be placed adjacent to n-type (NMOS) transistors. More specifically, a set of two or more PMOS transistors may be placed near a set of two or more NMOS transistors. In some circumstances, current may leak from the PMOS transistor, through the n-well, through the p-well, and to the NMOS transistor. This can cause a variety of issues such as latch-up. Latch-up is a type of short circuit that can occur in an integrated circuit. It involves the inadvertent creation of a low-impedance path through the integrated circuit. This triggers a parasitic structure which disrupts proper functioning of the circuit. Latch-up may result from two different types of transistors placed next to each other, i.e., a PMOS next to an NMOS transistor. This forms a PNP structure. Due to the finFET structure, particularly as fin widths become smaller, the dopant under the channel might be lost during a latch-up. Device leakage from subthreshold channel might also increase due to anti-punch through dopant being lost.

(9) According to principles described herein, a semiconductor device includes a first structure and a second structure. The first structure may be, for example, a pair of NMOS fin structures. The second structure may be, for example, a pair of PMOS fin structures. The top surface of the semiconductor bulk between the pair of fin structures of the first structure is higher than the top surface of the semiconductor structure between the first and second structures. Similarly, the top surface of the semiconductor bulk between the pair of fin structures of the second structure is higher than the top surface of the semiconductor structure between the first and second structures. This increases the physical distance through the semiconductor bulk between the NMOS and PMOS devices of the first and second structures. This reduces the potential current path between the two and reduces the chances that a parasitic structure associated with latch-up will occur.

(10) FIGS. 1A, 1B, 1C, 1D, 1E, and 1F illustrate a process for fabricating finFET structures with a crown bulk. FIG. 1A illustrates a substrate with a number of layers, including the well layers **102**, **104**, a semiconductor layer **106**, a hardmask layer **108**, and a patterned resist layer **110**. The well layers **102**, **104** may be doped portions of a semiconductor substrate. For example, the substrate may be a silicon substrate. The semiconductor substrate may be part of a silicon wafer. Other semiconductor materials are contemplated. The substrate may have a first region **102** and a second region **104**. The first region **102** may be, for example, an n-well. The second region **104** may be, for example, a p-well. An n-well is doped with an n-type dopant and the p-well is doped with a p-type dopant. Various dopant processes may be used such as implanting processes. In one example, the first region **102** may be covered with a resist material while the second region is doped. Additionally, the second region **104** may be covered with a resist material while the first region **102** is doped.

(11) In both regions **102**, **104**, the upper portions **102a**, **104a** may be doped with a higher doping concentration to form an anti-punch-through layer. The anti-punch-through layers **102a**, **104a** may extend from the top surface of the well regions **102**, **104** at line **101** to line **103**. The thickness **105** of the anti-punch-through layers **102a**, **104a** may be about 15-25 nanometers. Other sizes are contemplated. The anti-punch-through layers **102a**, **104a** are directly beneath the channel regions which will be formed within the semiconductor layer **106**.

(12) Anti-punch-through layers **102a**, **104a** provide various benefits. As transistors are formed with smaller sizes, the channels of such devices also become smaller. Smaller channels may present a variety of issues, which are often referred to as the short channel effect. For example, a short channel may allow for current to inadvertently flow between the source and the drain based on voltage differential between the source and drain. To avoid this issue, anti-punch-through features **102a**, **104a** can be formed at or near the bottom of a channel.

(13) To form the anti-punch-through layers in the n-well region **102**, an implanting process can be used. The implanting process can be tuned such that the anti-punch-through features are formed at a particular depth below the surface of the n-well **102**. In one example, the anti-punch-through

features **102a** are formed at a depth such that a bottom of the anti-punch-through features **102a** are at about 15-25 nanometers below the surface. This can be done by adjusting the electric field used in the ion implanting process. Ion implantation utilizes an electric field to accelerate ions towards a surface. By setting the strength of the electric field appropriately, the ions can lodge near a specific point below the surface. The implanting process implants a n-type dopant but at a higher concentration than the rest of the features **102**, which are already doped with an n-type dopant. The implanting process is such that the anti-punch-through layer is at a substantially uniform depth. In some examples, a Rapid Thermal Annealing (RTA) process is performed after the implanting process. An RTA process involves exposing the substrate to high temperatures.

(14) To form the anti-punch-through layers in the p-well region **104**, an implanting process can be used. The implanting process can be tuned such that the anti-punch-through features are formed at a particular depth below the surface of the p-well **104**. In one example, the anti-punch-through features **104a** are formed at a depth such that a bottom of the anti-punch-through features **104a** are at about 15-25 nanometers below the surface. The implanting process implants a p-type dopant but at a higher concentration than the rest of the features **104**, which are already doped with a p-type dopant. The implanting process is such that the anti-punch-through layer **104a** is at a substantially uniform depth. In some examples, a Rapid Thermal Annealing (RTA) process is performed after the implanting process. An RTA process involves exposing the substrate to high temperatures.

(15) The semiconductor layer **106** may be formed through an epitaxial growth process. The epitaxial growth process is used to form crystal structures on underlying crystal structures. In this case, the semiconductor layer **106** is grown onto the doped well regions **102**, **104**. The semiconductor layer **106** may be, for example, a silicon layer. Other semiconductor materials may be used as well. In some examples, the top 10 angstroms of the silicon layer may be formed in a separate epitaxial process. In other words, the first portion of the semiconductor layer **106** may be formed in a first epitaxial process, and the last 10 angstroms of the semiconductor layer **106** may be formed in a second, separate epitaxial process. This may be done to help control etching during subsequent steps.

(16) The hardmask layer **108** is deposited on the semiconductor layer **106**. The hardmask layer **108** is used to pattern the semiconductor layer **106** and the well layers **102**, **104** to form fin structures. The hardmask layer **108** may include at least one of silicon oxide (SiO₂), silicon nitride (SiN), silicon carbide (SiC), silicon oxynitride (SiON), silicon oxycarbide (SiOCN), hafnium oxide (HfO₂), aluminum oxide (Al₂O₃), and zirconium oxide (ZrO₂). Other materials are contemplated. In some examples, the hardmask layer **108** may include several sublayers. For example, the hardmask layer may include a first oxide layer approximately 25 angstroms thick. In some examples, the first oxide layer may be within a range of about 20-30 angstroms thick. On top of that, there may be a silicon nitride layer approximately 260 angstroms thick (or within a range of about 220-300 angstroms thick). On top of that, there may be another oxide layer approximately 450 angstroms thick (or within a range of about 400-500 angstroms thick).

(17) A photoresist **110** may be placed on top of the hardmask layer **108**. The photoresist **110** may be used to photolithographically pattern the hardmask layer **108**. For example, the photoresist **110** may be exposed to a light source through a photomask. The photoresist may then be developed such that the portions of the photoresist other than where fin structures are to be formed are removed.

(18) FIG. 1B illustrates a patterning process **112** in which the hardmask layer **108** is patterned to form a patterned hardmask layer **114**. The patterning process **112** may include exposing and developing the photoresist, and then etching the hard mask through the patterned photoresist. This will transfer the pattern of the photoresist to the hardmask layer **108** to form the patterned hardmask layer **114**. The patterned hardmask layer **114** includes a number of features that correspond to locations at which fin structures are to be formed. After the patterned hardmask layer **114** is formed, the semiconductor layer **106** is ready for etching.

(19) FIG. 1C illustrates an etching process **116**. The etching process **116** may be an anisotropic

etching process such as a dry etching process. The etching process **116** transfers the pattern of the patterned hardmask layer **114** to the semiconductor layer **106** and the well regions **102**, **104**. Thus, the etching process **116** forms a pair of fin structures **118a**, **118b** in the n-well region **102** and forms a pair of fin structures **118c**, **118d** in the p-well region **104**.

(20) The etching process etches fully through the semiconductor layer **106** and partially through the well regions **102**, **104**. The etching process **116** may be controlled to etch the well regions to a depth **107** that is twice the depth of the anti-punch-through layers **102a**, **104a**. In some examples, the etching process **116** may etch to a depth **109** of about 40 nanometers.

(21) The etching process **116** may be a dry etching process. Dry etching processes remove material by exposing the material to a bombardment of ions. This is done using plasma of reactive gases such as oxygen or chlorine with the addition of nitrogen or argon. The ions thus dislodge portions of the material from the exposed regions. Dry etching processes are generally anisotropic, which means that they etch primarily in one direction.

(22) FIG. 1D illustrates the formation of a patterned photoresist **126** on the workpiece. Specifically, a photoresist layer may be initially placed on the workpiece through a spin coating process. After the spin coating process, the photoresist layer exists as a single, continuous layer across the semiconductor wafer. Then, the photoresist layer is exposed to a light source through a photomask. After the exposure, the photoresist is developed. After development, the photoresist layer becomes the patterned photoresist layer **126**. The patterned photoresist layer covers the space between fin structure **118a** and **118b** as well as the space between fin structure **118c** and **118d**. However, the space outside those fin structures is exposed, as shown in FIG. 1D.

(23) FIG. 1E illustrates a second etching process **128** to further etch the n-well **102** and p-well **104** regions. This etching process **128** may be, for example, an anisotropic process such as a dry etching process. The etching process **128** may remove n-well **102** and p-well **104** regions to a particular depth at line **111**. The etching process **128** may remove approximately half as much of the n-well and p-well regions as the previous etching process **116** removed. For example, the etching process **128** may remove approximately 15-25 nanometers worth of material. Thus, the distance **113** between line **111** and line **107** may be approximately 15-25 nanometers.

(24) The etching process **128** may be a dry etching process. Dry etching processes remove material by exposing the material to a bombardment of ions. This is done using plasma of reactive gases such as oxygen or chlorine with the addition of nitrogen or argon. The ions thus dislodge portions of the material from the exposed regions. Dry etching processes are generally anisotropic, which means that they etch primarily in one direction.

(25) FIG. 1F illustrates a removal process **134** to remove the patterned photoresist layer **126**. After the patterned photoresist layer **126** is removed, subsequent fabrication steps may be performed to build the transistor on the fin structure. Specifically, a dummy gate may be formed over the fin structures in both the n-well and p-well regions. Source/drain regions may then be formed within the fin structure on both sides of the dummy gate. The dummy gate may then be replaced with a metal gate. In some examples, an isolation structure may be formed around the fin structures. The isolation structure may include, for example, an Interlayer Dielectric Layer (ILD) or a Shallow Trench Isolation (STI) structure. Either the ILD or the STI may include a low-k dielectric material. For example, the isolation structure may include one of silicon nitride (SiN), silicon carbon nitride (SiCN), or silicon oxycarbide (SiOCN). Other materials may be used as well.

(26) Using the techniques described herein, the top surface **136** of the n-well region **102** between the pair of fin structures **118a**, **118b** is higher than the top surface **140** between the two pairs of fin structures. Similarly, the top surface **138** of the p-well region **104** between the pair of fin structures **118c**, **118d** is higher than the top surface **140** between the two pairs of fin structures. This increases the path **142** between the two surfaces **136**, **138** and thus reduces the change of leakage current and latch-up.

(27) FIG. 2 is a diagram showing an illustrative image of finFET structures with a crown bulk. FIG.

2 illustrates a diagram that more closely resembles the actual shape of the devices described herein. FIG. 2 also illustrates the source/drain features **202** formed within the fin structures **118a**, **118b**, **118c**, **118d**. Specifically, line **101** represents the boundary between the channel/source/drain portion of the fin structures **118a**, **118b**, **118c**, **118d** and the top ends of the n-well **102** and p-well **104** region. At the top ends of the n-well and p-well regions **102**, **104** is the anti-punch-through layer **102a**, **104a**, which has a higher concentration of dopant than the lower portions of the fin structures **118a**, **118b**, **118c**, **118d**. The anti-punch-through layer **102a**, **104a** extends down to approximately line **103**. The thickness **105** of the anti-punch-through layers **102a**, **104a** may be approximately 15-25 nanometers. At line **107** is the top surfaces **136**, **138** of the bulk semiconductor substrate. Line **107** may be about twice the depth as line **103**. In some examples, line **107** may be at a depth **109** of about 40 nanometers. The distance **113** between the top surfaces **136**, **138** and the top surface **140** between the two pairs of fin structures may be about $\frac{1}{3}$ or 33% of the distance between line **111** and line **101**. Said differently, the distance **113** divided by (distance **113** plus distance **109**) is approximately 0.33. The ratios described herein are approximate and may vary within a range of about 0-15% from the stated distances and/or ratios.

(28) As a result of the processes described with the text accompanying FIGS. 1A-1F, a semiconductor device **200** is formed. The semiconductor device **200** includes a first structure **201** and a second structure **203**. The first structure **201** may be, for example, a pair of NMOS finFET transistors. FIG. 2 illustrates a cut along the source/drain features **202** and thus does not show the gate structure associated with the NMOS transistors. The first structure includes two fin structures **118a**, **118b**. The bottom portions of the fin structures, which correspond to the well region **102**, are doped with an n-type dopant. The upper ends of the bottom portions include an anti-punch-through layer **102a**. The anti-punch-through layer **102a** has a higher doping concentration than the lower portions of the n-well **102** (i.e., below line **103**). The first structure also includes a top surface **136** extending between fin structure **118a** and **118b**.

(29) The second structure **203** includes two fin structures **118c**, **118d**. The bottom portions of the fin structures, which correspond to the well region **104**, are doped with a p-type dopant. The upper ends of the bottom portions include an anti-punch-through layer **104a**. The anti-punch-through layer **104a** has a higher doping concentration than the lower portions of the p-well **104** (i.e., below line **103**). The second structure **203** also includes a top surface **138** extending between fin structure **118c** and **118d**. The surfaces **136** and **138** are higher than the surface **140** that extends between the first structure **201** and the second structure **203**.

(30) The ratios described above provide various advantages and benefits. Specifically, as mentioned above, having the higher surfaces **136**, **138** between fin structures than the surface **140** between the pairs of fin structures increases the distance between the surfaces **136**, **138** and reduces the chance of a latch-up occurring. Furthermore, by having some space between the top surfaces **136**, **138** and the anti-punch-through layers **102a**, **104a** (i.e., the distance between lines **103** and **107**) the anti-punch-through layer is able to work more effectively while still reducing the chances of latch-up. In other words, if the surfaces **136**, **138** were too close to the anti-punch-through layers **102a**, **104a**, then the benefits of reducing latch-up would be reduced because the higher dopant concentration of the anti-punch-through layers **102a**, **104a** would be partially diffused into the semiconductor substrate near the top surfaces **136**, **138**. Conversely, if the surfaces **136**, **138** are farther away from the anti-punch-through layers **102a**, **104a**, then they would be too close to the surface **140** (line **111**) and the advantages of avoiding latch-up would be reduced.

(31) FIG. 3 is a flowchart showing an illustrative method for forming finFET structures with a crown bulk. According to the present example, the method **300** includes a process **302** for forming a semiconductor layer on an n-well (e.g., **102**) and a p-well (e.g., **104**). The n-well and p-well may also be referred to as a first region and a second region. As mentioned above, an n-well is doped with an n-type dopant and the p-well is doped with a p-type dopant. Various doping processes may be used such as implanting processes. In one example, to form the n-well and p-well, the first

region may be covered with a resist material while the second region is doped. Additionally, the second region may be covered with a resist material while the first region is doped.

(32) The n-well and p-well may each include an anti-punch-through layer (e.g., **102a**, **104a**) in the upper portions thereof. The anti-punch-through layers have higher doping concentrations of their respective dopant types than the lower portions of the n-well and p-well. The anti-punch-through layers may extend from the top surface of the well regions to a depth of about 20 nanometers, in some examples. Other sizes are contemplated. The anti-punch-through layers may be directly beneath the channel regions which will be formed within the semiconductor layer (e.g., **106**).

(33) To form the anti-punch-through layers in the n-well and p-well regions, an implanting process can be used. The implanting process can be tuned such that the anti-punch-through features are formed at a particular depth below the surface of the well regions **102**, **104**. In one example, the anti-punch-through layers **102a**, **104a** are formed at a depth such that a bottom of the anti-punch-through layers **102a**, **104a** are at about 15-25 nanometers below the surface. This can be done by adjusting the electric field used in the ion implanting process. Ion implantation utilizes an electric field to accelerate ions towards a surface. By setting the strength of the electric field appropriately, the ions can lodge near a specific point below the surface. The implanting process implants a n-type dopant but at a higher concentration than the rest of the well regions **102**, **104**, which are already doped with an n-type dopant. The implanting process is such that the anti-punch-through layer is at a substantially uniform depth. In some examples, a Rapid Thermal Annealing (RTA) process is performed after the implanting process. An RTA process involves exposing the substrate to high temperatures.

(34) The semiconductor layer may be formed through an epitaxial growth process. The epitaxial growth process is used to form crystal structures on underlying crystal structures. In this case, the semiconductor layer is grown onto the doped well regions. The semiconductor layer may be, for example, a silicon layer. Other semiconductor materials may be used as well. In some examples, the top 8-12 angstroms of the silicon layer may be formed in a separate epitaxial process. In other words, the first portion of the semiconductor layer may be formed in a first epitaxial process, and the last 8-12 angstroms of the semiconductor layer may be formed in a second, separate epitaxial process. This may be done to help control etching during subsequent steps.

(35) According to the present example, the method **300** further includes a process **304** for forming a patterned hardmask layer (e.g., **108**) on the semiconductor layer. The hardmask layer may be deposited on the semiconductor layer. The hardmask layer **108** may be used to pattern the semiconductor layer as well as the well layers **102**, **104** to form fin structures (e.g., **118a**, **118b**, **118c**, **118d**). The hardmask layer may include at least one of silicon oxide (SiO₂), silicon nitride (SiN), silicon carbide (SiC), silicon oxynitride (SiON), silicon oxycarbide (SiOCN), hafnium oxide (HfO₂), aluminum oxide (Al₂O₃), and zirconium oxide (ZrO₂). Other materials are contemplated. In some examples, the hardmask layer may include several sublayers. For example, the hardmask layer may include a first oxide layer approximately 20-30 angstroms thick. On top of that, there may be a silicon nitride layer approximately 200-300 angstroms thick. On top of that, there may be another oxide layer approximately 400-500 angstroms thick. The hardmask layer may be patterned using a photolithographic process. For example, a photoresist may be placed on top of the hardmask layer. The photoresist may be exposed to a light source through a photomask. The photoresist may then be developed such that the portions of the photoresist other than where fin structures are to be formed are removed.

(36) According to the present example, the method **300** further includes a process **306** for, through the hard mask, in a first etching process (e.g., **116**), etching through the semiconductor layer and partially through the n-well and p-well to a first depth (e.g., **107**). The etching process may be an anisotropic etching process such as a dry etching process. The etching process transfers the pattern of the patterned hardmask layer to the semiconductor layer and the well regions. Thus, the etching process forms a pair of fin structures (e.g., **118a**, **118b**) in the n-well region and forms a pair of fin

structures (e.g., **118c**, **118d**) in the p-well region. The etching process etches fully through the semiconductor layer and partially through the well regions. The etching process may be controlled to etch the well regions to a depth (e.g., **109**) that is twice the depth of the anti-punch-through layers. In some examples, the etching process may etch to a depth of about 40 nanometers. The etching process may be a dry etching process.

(37) According to one example, the method **300** further includes a process **308** for forming a patterned photoresist layer such that photoresist material (e.g., **126**) is positioned between sets of fin structures. The photoresist layer may be initially placed on the workpiece through a spin coating process. After the spin coating process, the photoresist layer exists as a single, continuous layer across the semiconductor wafer. Then, the photoresist layer is exposed to a light source through a photomask. After the exposure, the photoresist is developed. After development, the photoresist layer becomes a patterned photoresist layer **126** that covers the space between fin structure **118a** and **118b** as well as the space between fin structure **118c** and **118d**. However, the space between those pairs of fin structures is exposed. Specifically, the space above the junction between the n-well and the p-well is exposed.

(38) According to the present example, the method **300** further includes a process **310** for, in a second etching process (e.g., **128**), further etching through the n-well and p-well to a second depth (e.g., **111**) in regions exposed through the photoresist layer. This second etching process may be, for example, an anisotropic process such as a dry etching process. The etching process may remove n-well and p-well regions to a particular depth. The etching process **128** may remove approximately half as much of the n-well and p-well regions as the previous etching process **116** removed. For example, the etching process may remove approximately 15-25 nanometers worth of material. The etching process may be a dry etching process.

(39) In some examples, after the second etching process, the photoresist layer may be removed. After the patterned photoresist layer **126** is removed, subsequent fabrication steps may be performed to build the transistor on the fin structure. Specifically, a dummy gate may be formed over the fin structures in both the n-well and p-well regions. Source/drain regions may then be formed within the fin structure on both sides of the dummy gate. The dummy gate may then be replaced with a metal gate. In some examples, an isolation structure may be formed around the fin structures. The isolation structure may include, for example, an Interlayer Dielectric Layer (ILD) or a Shallow Trench Isolation (STI) structure. Either the ILD or the STI may include a low-k dielectric material. For example, the isolation structure may include one of silicon nitride (SiN), silicon carbon nitride (SiCN), or silicon oxycarbide (SiOCN). Other materials may be used as well.

(40) By using principles described herein, finFET structures can be improved. For example, the diffusion area within the crown bulk may be reduced. Additionally, there is less leakage current between finFET devices. This reduces the likelihood that a latch-up will occur. And, the well resistance may be lowered as well, which can also reduce the likelihood of a latch-up. Both the leakage current and the well resistance may be reduced by 1 order of magnitude.

(41) According to one example, a semiconductor device includes a first structure having a first fin structure, a second fin structure wherein bottom portions of the first fin structure and the second fin structure are doped with an n-type dopant, and a first semiconductor surface extending between the first fin structure and the second fin structure. The device further includes a second structure having a third fin structure, a fourth fin structure wherein bottom portions of the third and fourth fin structures are doped with a p-type dopant, and a second semiconductor surface extending between the third structure and the fourth structure. The device further includes a third semiconductor surface extending between the first structure and the second structure, the third semiconductor surface being at a lower level than the first and second semiconductor surfaces.

(42) A device includes a semiconductor substrate having a first region and a second region. The device further includes a first pair of fin structures within the first region. The device further includes a second pair of fin structures within the second region. A top surface of the

semiconductor surface between fin structures within the first pair is higher than a top surface of the semiconductor surface between the first pair and the second pair.

(43) A method includes forming a semiconductor layer on an n-well and a p-well, forming a patterned hardmask layer on the semiconductor layer, through the hard mask, in a first etching process, etching through the semiconductor layer and partially through the n-well and p-well to a first depth, forming a patterned photoresist layer such that photoresist material is positioned between sets of fin structures, and in a second etching process, further etching through the n-well and p-well to a second depth in regions exposed through the photoresist layer.

(44) The foregoing outlines features of several embodiments so that those skilled in the art may better understand the aspects of the present disclosure. Those skilled in the art should appreciate that they may readily use the present disclosure as a basis for designing or modifying other processes and structures for carrying out the same purposes and/or achieving the same advantages of the embodiments introduced herein. Those skilled in the art should also realize that such equivalent constructions do not depart from the spirit and scope of the present disclosure, and that they may make various changes, substitutions, and alterations herein without departing from the spirit and scope of the present disclosure.

Claims

1. A method, comprising: forming an anti-punch-through layer over a first region and a second region of a substrate, wherein the first region and the second region of the substrate are of opposite conductive types; forming a semiconductor layer over the anti-punch-through layer; patterning the semiconductor layer and the anti-punch-through layer to form a first plurality of fins over the first region and a second plurality of fins over the second region; forming a patterned resist layer over the first plurality of fins and the second plurality of fins, wherein a portion of the substrate between the first plurality of fins and the second plurality of fins is exposed in an opening of the patterned resist layer; and recessing the portion of the substrate between the first plurality of fins and the second plurality of fins, such that a top surface of the portion of the substrate between the first plurality of fins and the second plurality of fins is below a first semiconductor surface extending within the first plurality of fins and a second semiconductor surface extending within the second plurality of fins.
2. The method of claim 1, further comprising: forming a first epitaxial feature atop the anti-punch-through layer in the first region; and forming a second epitaxial feature atop the anti-punch-through layer in the second region.
3. The method of claim 2, wherein a bottom surface of the first epitaxial feature directly interfaces the anti-punch-through layer, and a bottom surface of the second epitaxial feature directly interfaces the anti-punch-through layer.
4. The method of claim 1, wherein the patterning of the semiconductor layer and the anti-punch-through layer also recesses the substrate to form the first semiconductor surface in the first region and the second semiconductor surface in the second region.
5. The method of claim 4, wherein a vertical distance from a top surface of the anti-punch-through layer to the first semiconductor surface is about twice of a thickness of the anti-punch-through layer.
6. The method of claim 1, wherein after the recessing of the portion of the substrate between the first plurality of fins and the second plurality of fins, a vertical distance from the first semiconductor surface to the top surface of the portion of the substrate between the first plurality of fins and the second plurality of fins is about one third of a vertical distance from a top surface of the anti-punch-through layer to the top surface of the portion of the substrate between the first plurality of fins and the second plurality of fins.
7. The method of claim 1, wherein a thickness of the anti-punch-through layer is about 15 nm to

about 25 nm, and wherein after the recessing of the portion of the substrate between the first plurality of fins and the second plurality of fins, a vertical distance from the first semiconductor surface to the top surface of the portion of the substrate between the first plurality of fins and the second plurality of fins is about 15 nm to about 25 nm.

8. The method of claim 1, wherein the forming of the patterned resist layer includes: depositing a resist layer continuously extending from the first region to the second region, and patterning the resist layer to form the patterned resist layer, the patterned resist layer filling space among the first plurality of fins and space among the second plurality of fins.

9. The method of claim 1, wherein after the forming of the patterned resist layer and prior to the recessing of the portion of the substrate between the first plurality of fins and the second plurality of fins, a first sidewall of a first fin of the first plurality of fins and a second sidewall of a second fin of the first plurality of fins are exposed, the first sidewall facing the second plurality of fins, the second sidewall facing in a direction away from the second plurality of fins.

10. The method of claim 1, wherein the forming of the anti-punch-through layer includes doping a top portion of the substrate in the first region and the second region.

11. A method, comprising: forming an anti-punch-through layer over a substrate; epitaxially growing a semiconductor layer over the anti-punch-through layer; patterning the semiconductor layer, the anti-punch-through layer, and a top portion of the substrate, thereby forming at least a first fin and a second fin protruding from the substrate; depositing a resist layer filling a space between the first and second fins, wherein a first sidewall of the first fin facing away from the second fin and a second sidewall of the second fin facing away from the first fin are exposed through openings of the resist layer; recessing the substrate through the openings of the resist layer, thereby extending downwardly the first and second sidewalls; and after the recessing of the substrate, removing the resist layer.

12. The method of claim 11, further comprising: forming an epitaxial feature above the first and second fins, wherein the epitaxial feature continuously extends from a first position directly above the first fin to a second position directly above the second fin.

13. The method of claim 12, wherein the epitaxial feature is in direct contact with the anti-punch-through layer.

14. The method of claim 11, wherein after the removing of the resist layer, a vertical distance from a top surface of the anti-punch-through layer to a top surface of the substrate between the first and second fins is about twice of a thickness of the anti-punch-through layer.

15. The method of claim 11, wherein after the removing of the resist layer, a vertical distance from a top surface of the anti-punch-through layer to a top surface of the substrate directly interfacing the first and second sidewalls is about three times of a thickness of the anti-punch-through layer.

16. The method of claim 11, a thickness of the anti-punch-through layer ranges from about 15 nm to about 25 nm.

17. A method comprising: forming a semiconductor layer on an n-well and a p-well; forming an anti-punch-through layer at a top end of the n-well and the p-well and below the semiconductor layer; forming a patterned hardmask layer on the semiconductor layer; through the patterned hardmask layer, in a first etching process, etching through the semiconductor layer and partially through the n-well and the p-well to a first depth, thereby forming sets of fin structures; forming a patterned photoresist layer such that photoresist material of the patterned photoresist layer is positioned between the fin structures of a same set; and in a second etching process, further etching through the n-well and the p-well to a second depth in regions exposed through the patterned photoresist layer.

18. The method of claim 17, wherein the second depth is about 15 nm to about 25 nm deeper than the first depth.

19. The method of claim 17, wherein a thickness of the anti-punch-through layer ranges from about 15 nm to about 25 nm.

20. The method of claim 17, wherein the forming of the anti-punch-through layer includes doping the top end of the n-well and the p-well prior to the forming of the semiconductor layer.
